

Nermine Melki Amdouni

Junior AI / NLP Engineer, Applied Machine Learning & Full-Stack Delivery
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PROFILE

Junior AI/NLP engineer and soon-to-graduate engineering student with hands-on internship experience across the LLM pipeline: speech recognition, machine translation, prompt engineering, and OCR/LLM data extraction on real-world, multilingual datasets. Also skilled in full-stack development (Flask, React, Angular, Node.js). Native Arabic speaker, C1 English. Seeking an entry-level AI/NLP or data engineering role.

PROFESSIONAL EXPERIENCE

- Data Consulting Intern — EY AMC**

02/2025 – 07/2025

 - Engineered an end-to-end real-time speech-translation pipeline (ASR to MT to TTS) integrating speech recognition, machine translation, and speech synthesis across 3 languages, achieving sub-5-second latency.
 - Optimized prompt design and pipeline modules to reach 90% translation accuracy, validating results through WER, CER, and BLEU benchmarking against client requirements.
 - Delivered the project within an Agile framework, participating in daily stand-ups and adapting scope continuously to evolving client requirements.

Technologies: Python, Flask, LangChain, Whisper, MarianMT, OpenAI API, Git
- Data Analyst / Development Intern — NEXT STEP**

06/2024 – 08/2024

 - Built automated data collection and structuring pipelines for cloud platform performance data, standardizing unstructured logs into a centralized MongoDB store.
 - Developed a full-stack monitoring web application (React, Node.js, Flask) used by the team to track cloud performance data in real time.
 - Collaborated within a development team following Git-based version control best practices.

Technologies: React, Node.js, MongoDB, Flask, Git
- Developer / Data Scientist (Academic Collaboration) — Swiss Premium Negoce**

01/2024 – 05/2024

 - Delivered a full-stack OCR/LLM application (Angular, Flask, MongoDB) integrating a Mistral 7B model to automate structured data extraction from 1,577 records and 25+ multilingual penalty images, replacing manual data entry.
 - Built the underlying data cleaning and annotation pipeline used to train and deploy the OCR/LLM model.
 - Created interactive Power BI dashboards to track application performance, extraction reliability, and business metrics for stakeholder reporting.

Technologies: Angular, Flask, MongoDB, Mistral 7B, Power BI

ACADEMIC PROJECT

- PMBOK Knowledge Graph & Chatbot**
 - Mapped project management concepts in a Neo4j knowledge graph and implemented a LangChain RAG system for contextual Q&A via a web application.

EDUCATION

- Engineering Degree, IT (Minor in Data Science) — ESPRIT**

09/2022 – 11/2025

 - International program with a multicultural cohort, fully taught in English.
- Preparatory Institute for Engineering Studies of El Manar — IPEIEM**

09/2020 – 08/2022

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, SQL
AI / NLP / LLMs: LangChain, Prompt Engineering, RAG, Whisper, MarianMT, Mistral 7B, OpenAI API, NLP
Full-Stack Web Development: Flask, React, Node.js, Angular, REST APIs, JSON
Databases: MongoDB, MySQL, Oracle, Neo4j
Analysis & Visualization: Power BI, Streamlit, ETL Pipelines
Tools: Git

CERTIFICATIONS

- NVIDIA: Building RAG Agents with LLMs
- NVIDIA: Building Transformer-Based Natural Language Processing Applications
- Microsoft Azure: Introduction to Cloud Services (Microsoft-authorized)

ADDITIONAL EXPERIENCE

- Marketing Department Member, INCEPTUM Junior Enterprise (09/2023 – 11/2025): designed pitch decks and marketing reports supporting partnership proposals; directed content projects from brief to delivery.
- Freelance Client & Digital Engagement Management (12/2025 – present): managed direct client relationships and social media performance for a small business, using engagement analytics to adjust strategy.

LANGUAGES

Arabic: Native | English: C1 | French: B2 | German: A1 | Spanish: A2